

Face Filters (PySide6 + OpenCV)

Real-time webcam face filters with sunglasses and hat overlays. Uses Haar cascades for quick detection and MediaPipe Face Mesh for accurate landmark placement.

Requirements

- Python 3.11
- Webcam

Setup (Windows PowerShell)

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
```

Or use the helper script:

```
scripts\create_venv.ps1
```

Run

```
.\.venv\Scripts\Activate.ps1
python -m face_graphics
```

Configuration

Set custom cascade paths if you want to override OpenCV defaults:

```
$env:FACE_CASCADE_PATH = "C:\\path\\to\\haarcascade_frontalface_alt.xml"
$env:EYE_CASCADE_PATH = "C:\\path\\to\\haarcascade_eye_tree_eyeglasses.xml"
```

The MediaPipe landmarker model is downloaded automatically on first run to:

```
models/face_landmarker.task
```

If you want a custom location, set:

```
$env:FACE_LANDMARKER_PATH = "C:\\path\\to\\face_landmarker.task"
```

Hand gesture rotation uses a separate model downloaded automatically to:

```
models/hand_landmarker.task
```

Override the path if needed:

```
$env:HAND_LANDMARKER_PATH = "C:\\path\\to\\hand_landmarker.task"
```